Department of Computer Science and Engineering
Academic Year 2021– 2022(Even Semester)

Degree, Semester & Branch: B.E – II Semester CSE
Course Code & Title: CS3251 Programming in C
Name of the Faculty member (s): Ms.G.Sakthi Priya, AP/CSE

Innovative Practice Description

- **Unit / Topic: Unit I / Decision and Iteration Statements**
- **Course Outcome:** CO1
- **Topic Learning Outcome:** TLO2, 3
- **Activity Chosen: Code Debugging**
- **Justification:**
  Erroneous code fails to provide expected output. Debugging is testing a program to find the cause of error. Code debugging activity helps the students to write error free code and to remember the syntax of decision-making statements
- **Time Allotted for the Activity:** 10 Minutes
- **Details of the Implementation:**
  Materials Required:
  - Small Game Board to display the code
  - Debugging Worksheet to spot the error in the code

  Activity:
  - Code with Errors were displayed to the students as shown in Fig 1.
  - Students were asked to step through the existing code to identify errors including missing blocks, extra blocks, spell check and syntax errors.
  - After identification of errors, they were asked to correct the program and write the error free code as shown in Fig 2.
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO9	PO10	PSO1
CO2	3	1	1	1	1	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO9	PO10	PSO1
Justification for correlation	It make the student able to apply the concept of syntax mapping to figure out the error in the	Students will be able to analyze the erroneous program by applying the syntax	Students will be able to provide solution for the problem by	As the student involved in the activity as an individual, their coding skill is	Students programming skill will be improved as they are debugging the code and documenting	The debugging skill earned through this activity helps the students in solving

	program	they learnt	correcting the error code	improved	the activity	various problems by writing error free code

- **Images / Screenshot of the practice:**

Debugging

```
#include <stdio.h>
int main()
{
   int age;
   printf("Enter your age?");
   scanf("%d",age)
   if(age>18)
   {
      printf("You are eligible to vote...");
   else
   {
      printf("Sorry ... you can't vote");
   }
   return 0;
}
```

Debugging

```
#include<stdio.h>
int main()  {
     int number=0;
     printf(enter a number:);
     scanf("%d",&number);
     if number==10 {
     printf("number is equals to 10");
     }
     else if number==50 {
     printf("number is equal to 50");
     }
     else if number==100{
     printf("number is equal to 100");
     }
     else{
     printf("number is not equal to 10, 50 or 100");
     }
     return 0;
}
```

Fig 1: Programming questions with errors

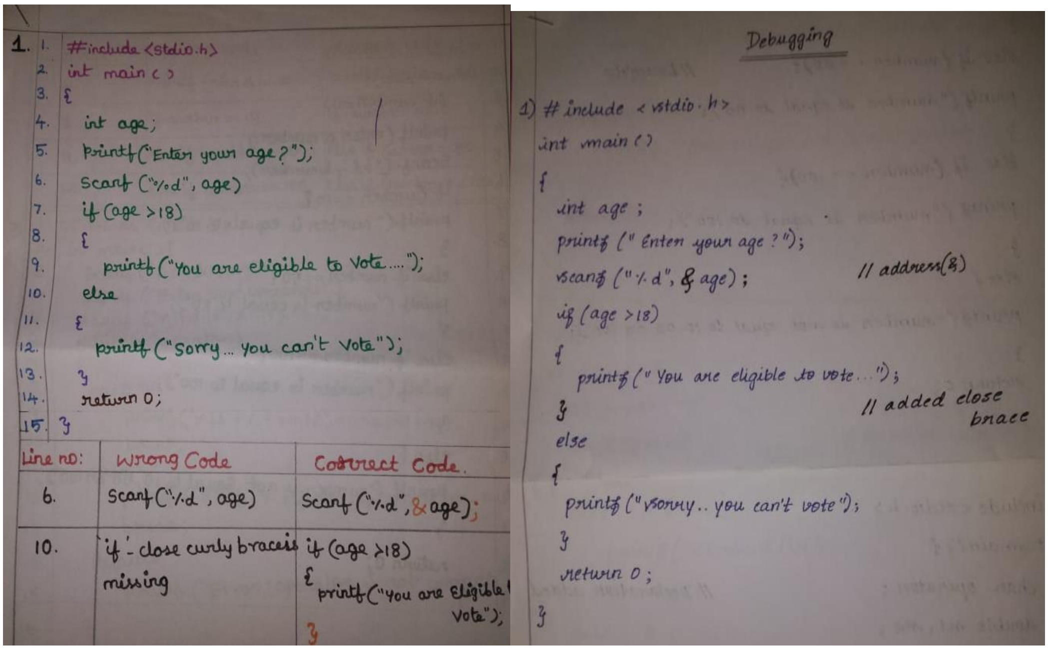

Fig 2: Corrected code submitted by the students

- **Reflective Critique:**
  - ***Feedback of practice from students and other stakeholders:***
    - The student felt the activity is interesting and helps them to remember the syntax.
    - They felt this will be very helpful when preparing for exams and practicing programs
  - ***Benefit of the practice:***
    - Students will be able to predict where a program will fail

- They can modify an existing program to solve error
- Students can identify an algorithm that is unsuccessful when the steps are out of order

❖ ***Challenges faced in implementation:***

- All the students were unable to complete the task within allotted time
- During the activity unable to cover the logical error type of program

References:

- https://code.org/curriculum/course1/5/Teacher
- https://www.canadalearningcode.ca/lessons/speaking-code-debugging-lesson-6-of-7/

Signature of Faculty Member **HOD**